Lab 2 - Windows Server & Active Directory Home Lab

1. Objective

The objective of this lab was to build and administer a small Windows domain environment using Windows Server 2022, Active Directory Domain Services (AD DS), DNS, organizational units, users, security groups, Group Policy, and a Windows 10 domain-joined client. The lab was designed to simulate common IT administration, networking, user management, permissions, and troubleshooting tasks in a small business environment.

Environment

- **Server:** Windows Server 2022 Standard Evaluation (Desktop Experience)
- **Client:** Windows 10 Pro
- **Virtualization:** Oracle VirtualBox
- **Domain:** zackilab.local
- **Domain Controller:** DC01
- **Server Host-Only IP:** 192.168.56.10
- **Client Host-Only IP:** 192.168.56.20

Tools & Technologies

- Windows Server 2022
- Active Directory Domain Services (AD DS)
- DNS
- Group Policy
- Active Directory Users and Computers
- Windows 10
- Oracle VirtualBox
- Command Prompt

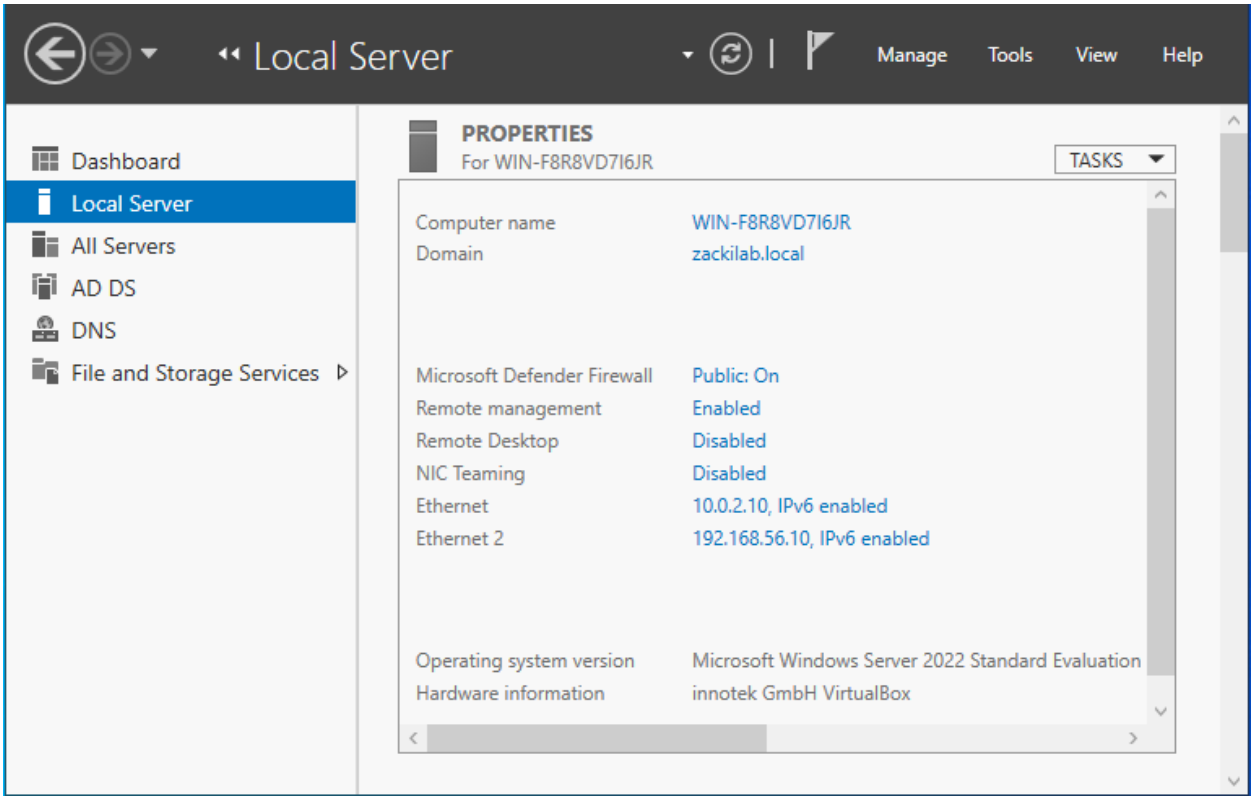
2. Windows Server Installation & Configuration

A Windows Server 2022 virtual machine was created in Oracle VirtualBox to serve as the domain controller for the lab environment. The server was configured with 4 GB of RAM, 2 CPU cores, and a 60 GB dynamically allocated virtual disk.

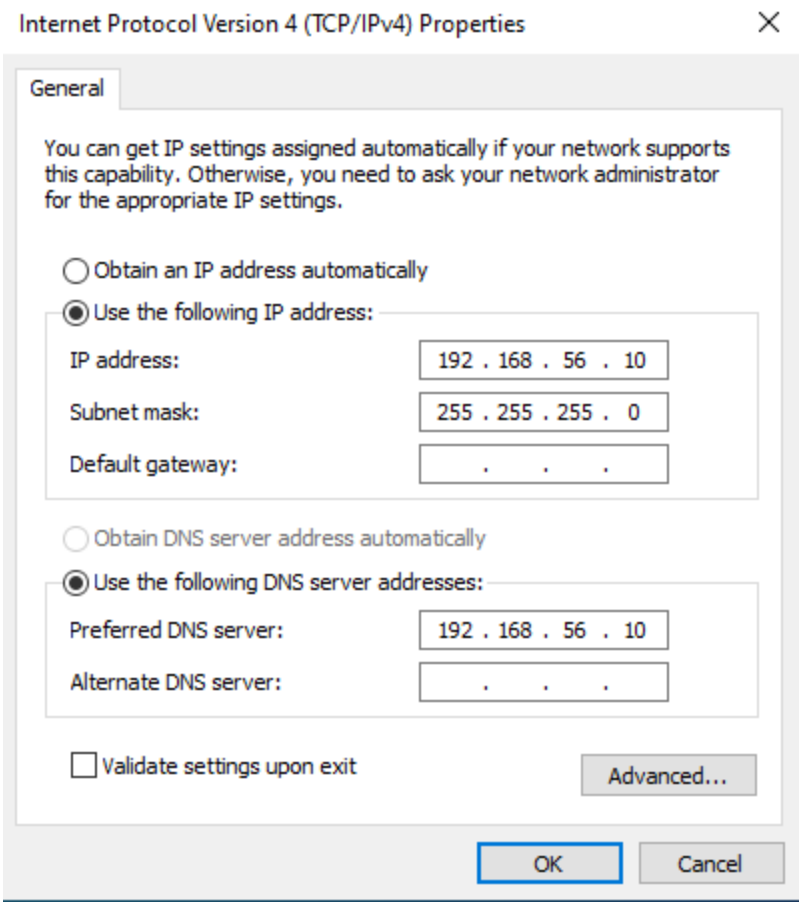
Windows Server 2022 Standard Evaluation with Desktop Experience was installed using a custom installation. After installation, the server was configured and prepared for use as the domain controller.

Evidence

Screenshot 01 — Server Configuration



Screenshot 02 — Server Static IP



3. Active Directory Domain Services & Domain Controller

Active Directory Domain Services was installed on the Windows Server. The server was then promoted to a domain controller and configured as the first domain controller in a new forest.

The domain created for the lab was:

zackilab.local

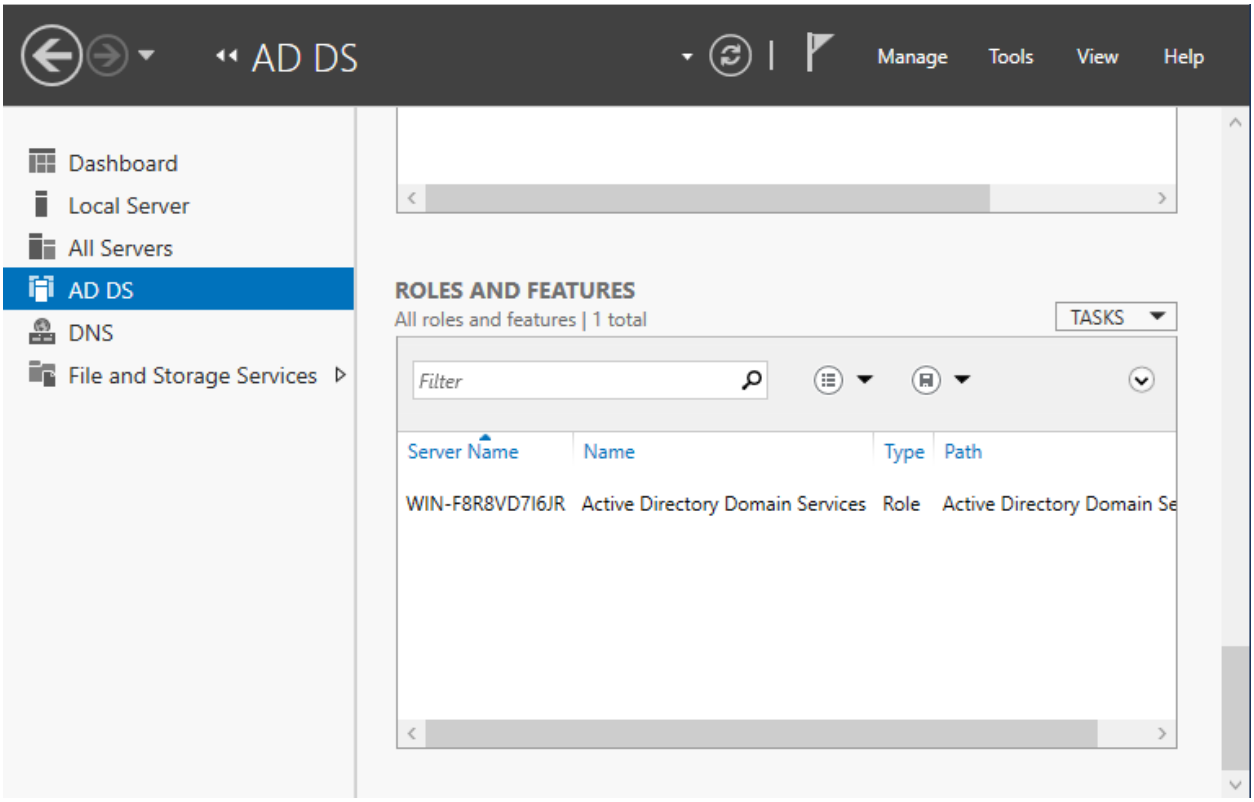
The NetBIOS domain name was configured as:

ZACKILAB

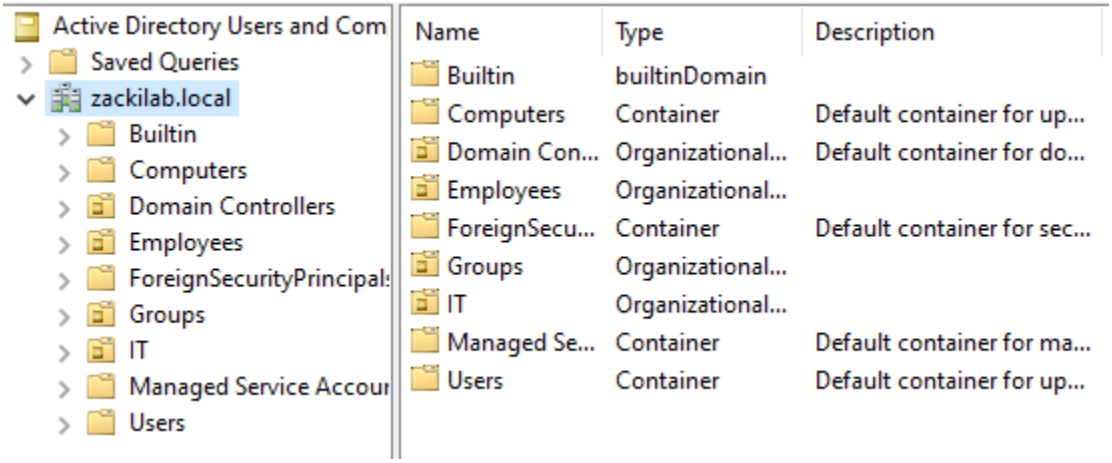
DNS and Global Catalog services were also configured during the domain controller promotion.

Evidence

Screenshot 03 — AD DS Installed



Screenshot 04 — Domain Controller



4. Organizational Units, Users & Security Groups

Organizational Units (OUs) were created to organize the Active Directory environment. The following OUs were configured:

- IT
- Employees
- Groups
- Computers

The Computers OU was automatically created by Active Directory.

Two domain user accounts were created inside the Employees OU:

- John Smith — `john.smith`
- Sarah Johnson — `sarah.johnson`

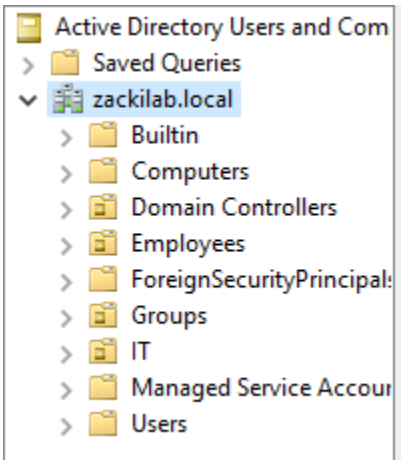
Two security groups were also created inside the Groups OU:

- IT-Support
- Employees

John Smith and Sarah Johnson were added to the Employees security group. John Smith was later added to the IT-Support group to test group-based permissions.

Evidence

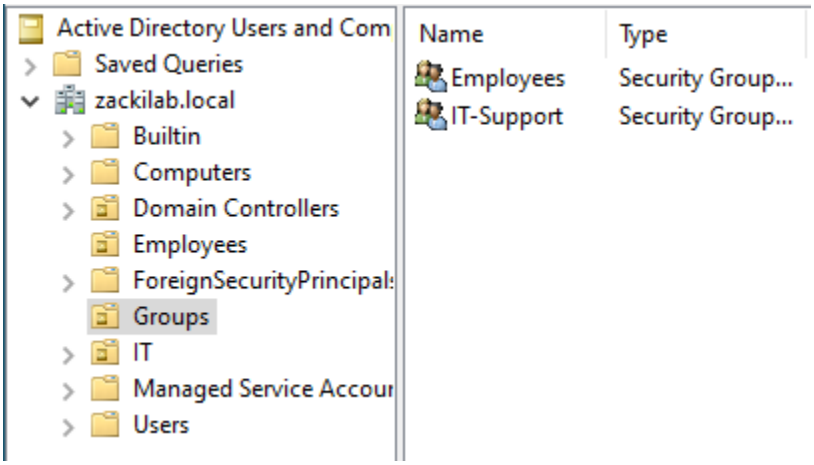
Screenshot 05 — Organizational Units



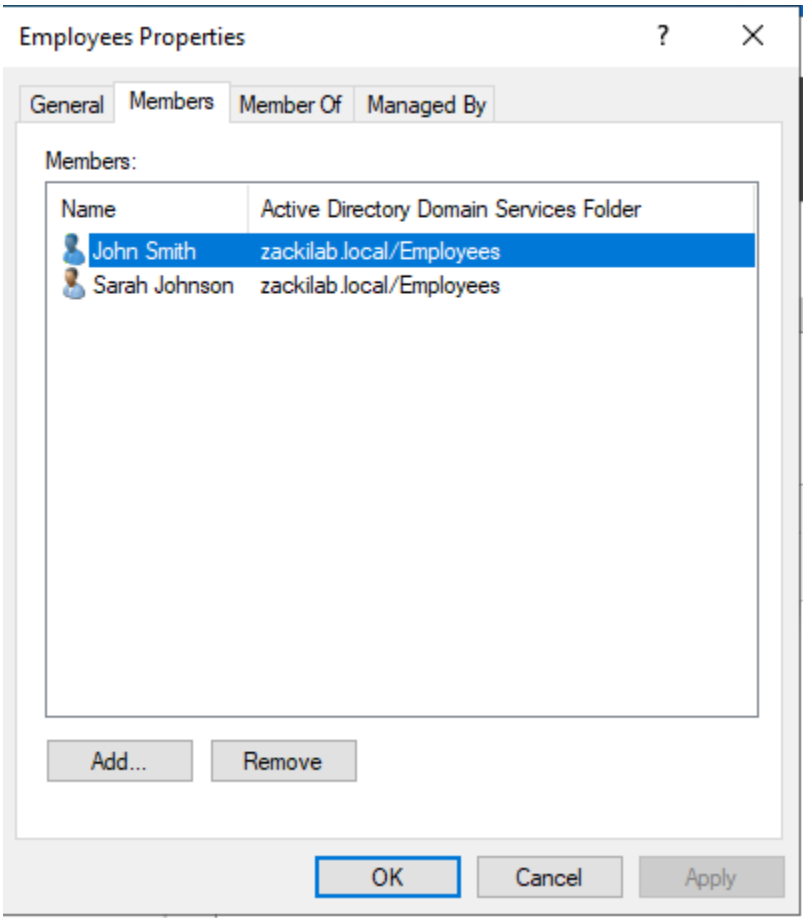
Screenshot 06 — Domain Users



Screenshot 07 — Security Groups



Screenshot 08 — Group Members



5. Virtual Network Configuration & Domain Joining

A private Host-only network was configured in VirtualBox to allow the Windows Server and Windows 10 client to communicate directly while remaining isolated from the physical network.

The server was configured with the Host-only address:

192.168.56.10

The Windows 10 client was configured with:

192.168.56.20

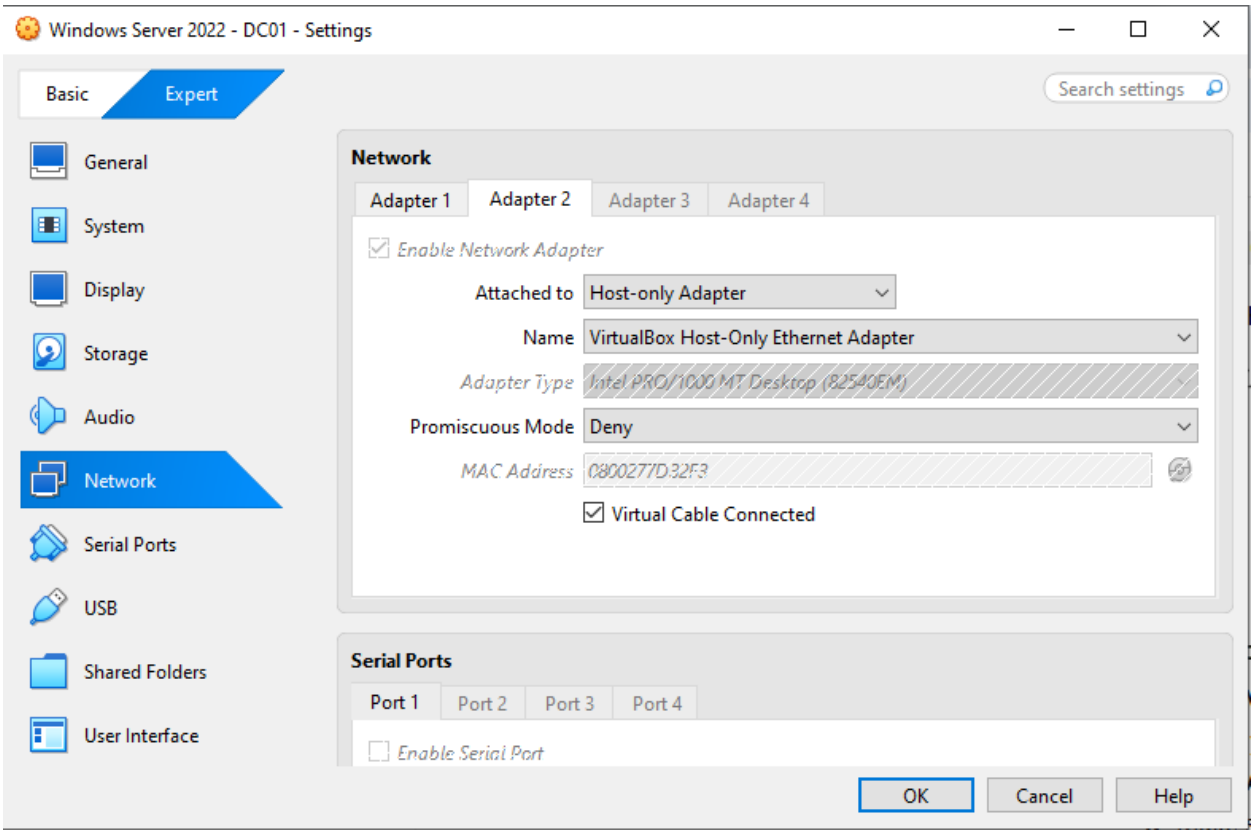
The server was configured as the client's DNS server using:

192.168.56.10

The Windows 10 client was successfully joined to the `zackilab.local` domain and was able to authenticate using a domain account.

Evidence

Screenshot 09 — Host-Only Network

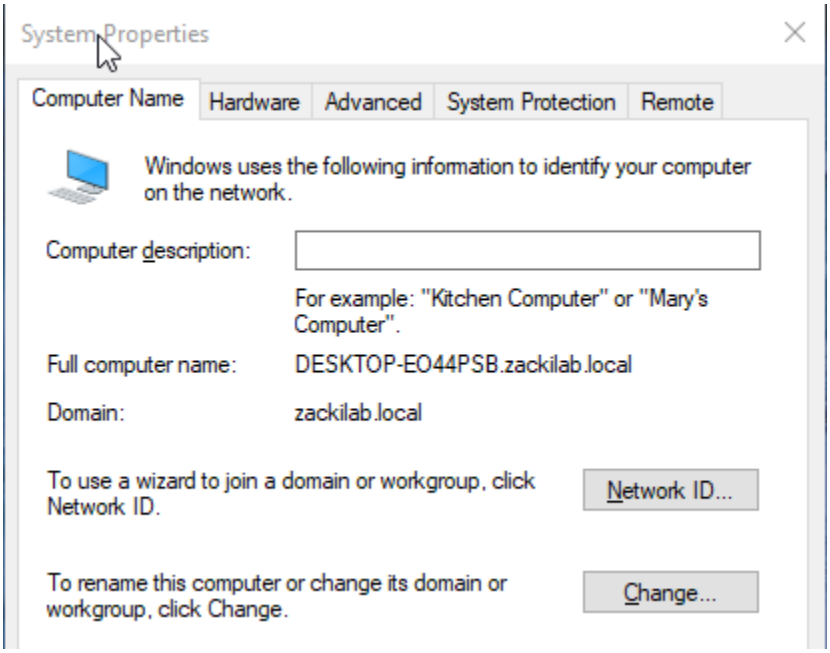


Screenshot 10 — Client IP Configuration

```
Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
   Physical Address. . . . . . . . . : 08-00-27-50-E5-9E
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::a13:6731:1ab3:22d%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.20(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 168296487
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-53-CA-23-08-00-27-E3-A9-2C
   DNS Servers . . . . . . . . . . . : 192.168.56.10
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Screenshot 11 — Domain Join



Screenshot 12 — Domain Login

```
C:\Users\john.smith>whoami
zackilab\john.smith
```

6. DNS & Network Troubleshooting

DNS configuration was tested using the Windows command line. During troubleshooting, the client initially attempted to use an external DNS server instead of the domain controller for domain name resolution.

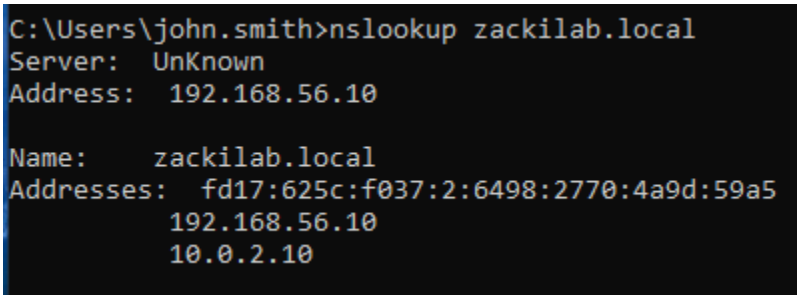
The DNS configuration was corrected so that the domain controller provided DNS resolution for the Active Directory environment. An alternate DNS configuration was also maintained for external name resolution.

The `nslookup` command was used to verify that `zackilab.local` successfully resolved through the domain controller.

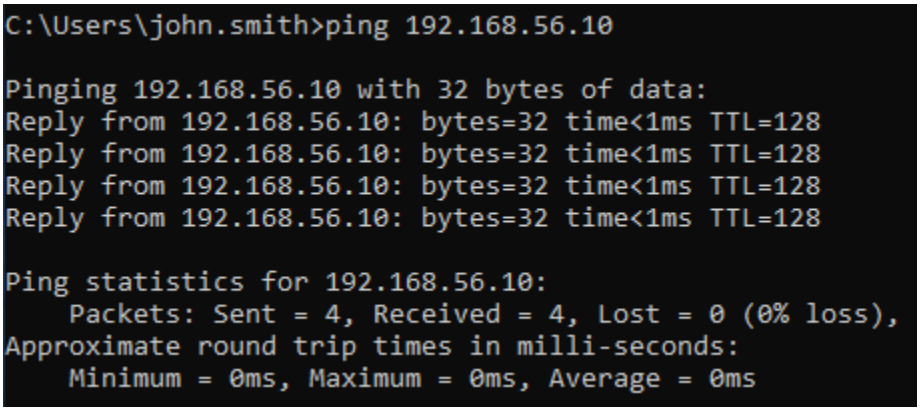
Network connectivity between the Windows client and domain controller was also verified using the `ping` command.

Evidence

Screenshot 13 — DNS Verification



Screenshot 14 — Network Connectivity



7. Shared Folder & Group-Based Permissions

A shared folder named `IT_Shared` was created on the domain controller to simulate a business network file share.

The folder was configured with shared permissions and NTFS security permissions. The `IT-Support` security group was given the appropriate permissions to access and modify the shared folder.

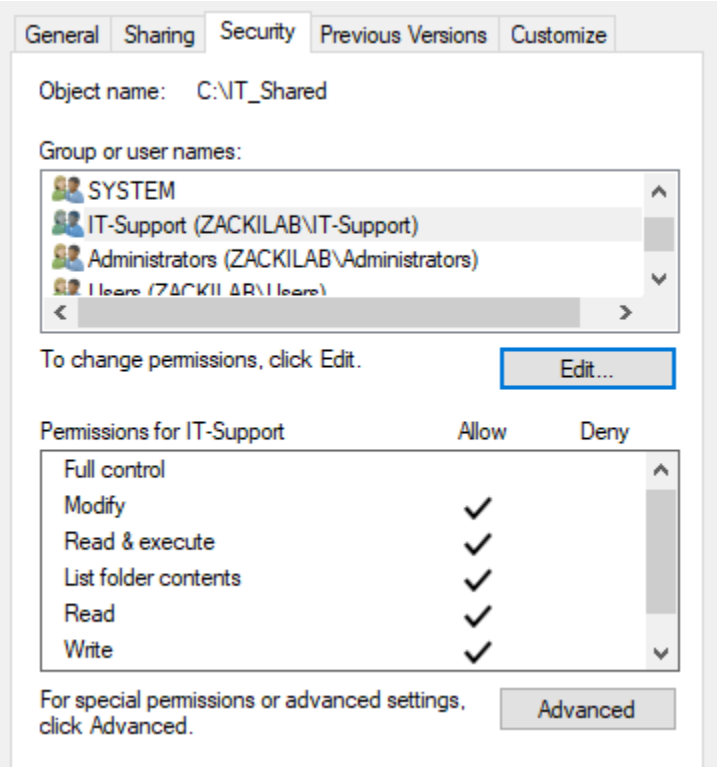
Testing was performed using the domain account `john.smith`.

When John Smith was not a member of the IT-Support group, access to the required shared-folder functionality was denied. After John Smith was added to the IT-Support group and logged back into the domain, he was able to access and modify the shared folder.

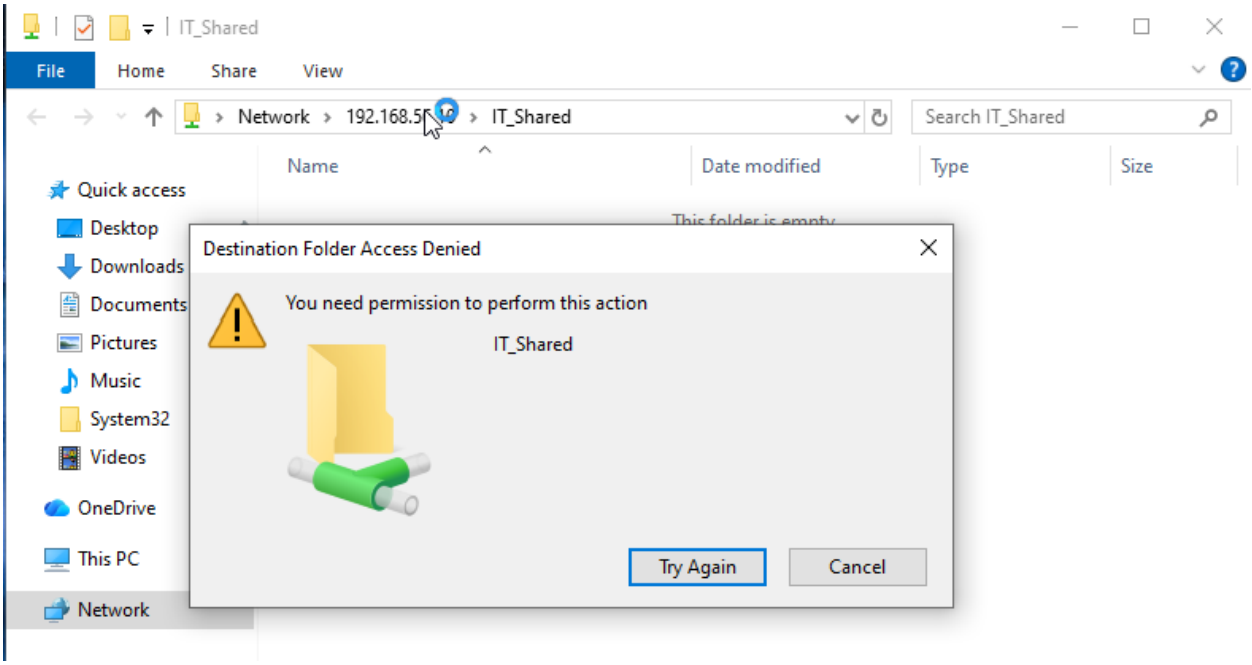
This demonstrated how Active Directory security groups can be used to control access to network resources.

Evidence

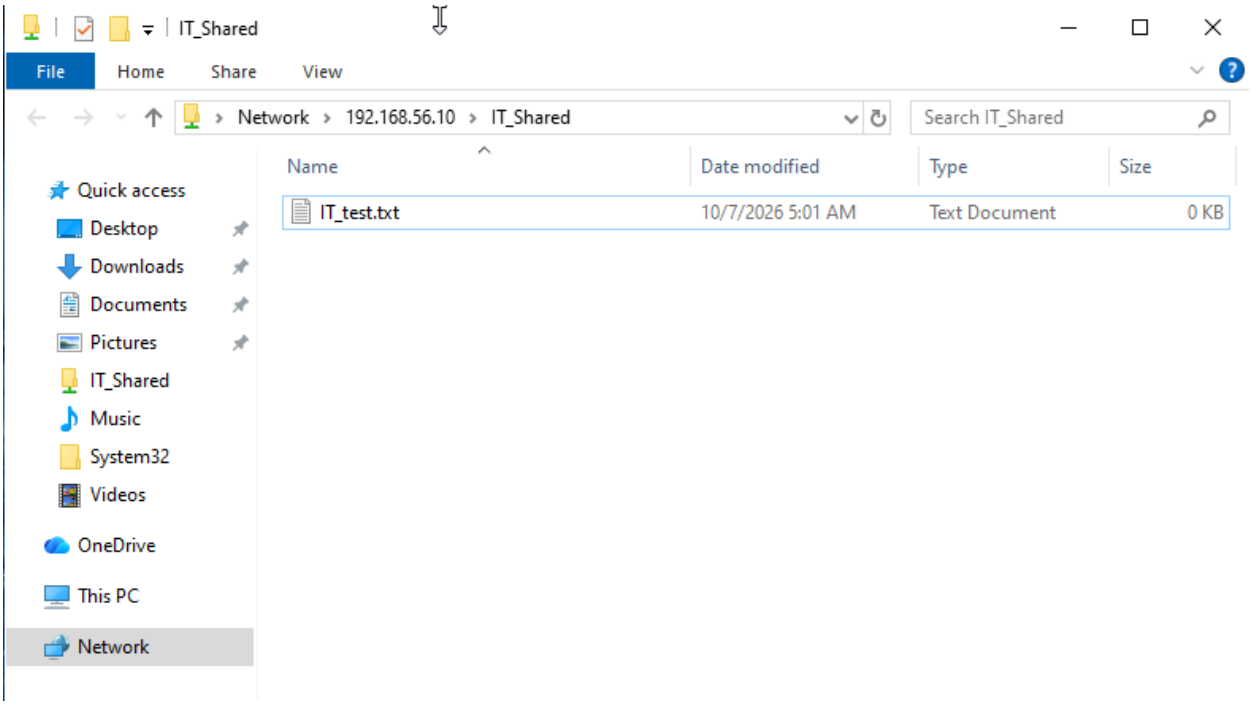
Screenshot 15 — Shared Folder Permissions



Screenshot 16 — Shared Folder Access Denied



Screenshot 17 — IT-Support Access



8. Group Policy

A Group Policy Object named `IT-Lab-Policy` was created and linked to the `zackilab.local` domain.

The policy was configured to demonstrate centralized security and configuration management through Group Policy.

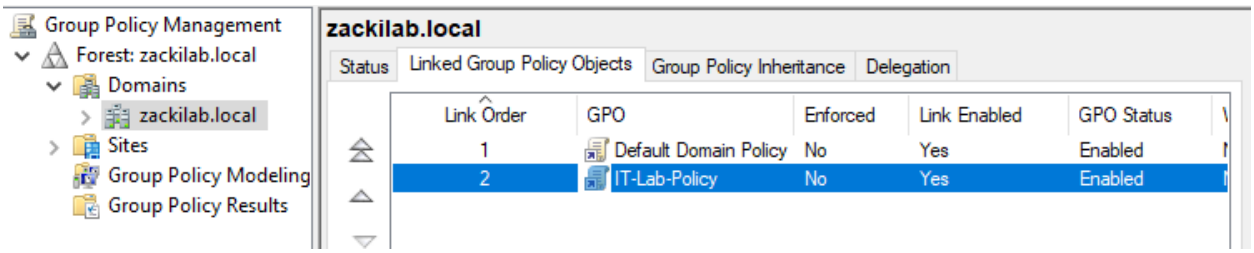
The Windows 10 client was then updated using:

```
gpupdate /force
```

This was used to verify that the client could receive and process domain-based Group Policy settings.

Evidence

Screenshot 18 — Group Policy



Screenshot 19 — Group Policy Applied

```
C:\Users\john.smith>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

9. Final Domain & Network Verification

The final stage of the lab consisted of verifying domain authentication, DNS resolution, shared-folder access, and network connectivity.

The `whoami` command confirmed that the client was authenticated using the domain account:

`zackilab\john.smith`

DNS resolution was verified using `nslookup`, and network connectivity to the domain controller was verified using `ping`.

The shared folder was also tested to confirm that the IT-Support security group provided the intended access.

Evidence

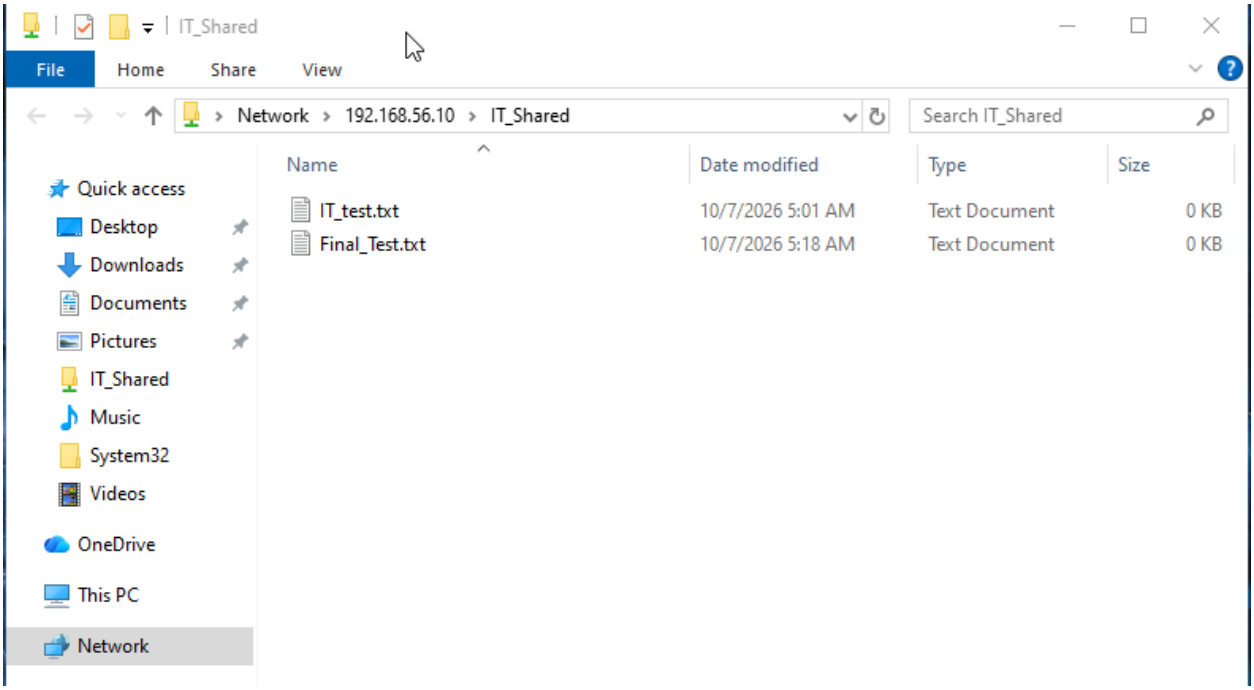
Screenshot 20 — Final Domain/DNS Verification

```
C:\Users\john.smith>whoami
zackilab\john.smith

C:\Users\john.smith>nslookup zackilab.local
Server:  UnKnown
Address:  192.168.56.10

Name:    zackilab.local
Addresses:  fd17:625c:f037:2:6498:2770:4a9d:59a5
          192.168.56.10
          10.0.2.10
```

Screenshot 21 — Final Shared Folder Test



Screenshot 22 — Final Network Verification

```
C:\Users\john.smith>ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\john.smith>ping zackilab.local

Pinging zackilab.local [192.168.56.10] with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

10. Conclusion & Skills Demonstrated

This lab successfully simulated a small business Windows domain environment using Windows Server 2022 and a Windows 10 domain-joined client.

The lab provided hands-on experience with:

- Windows Server administration
- Active Directory Domain Services
- Domain Controller configuration
- DNS configuration and troubleshooting
- Active Directory Users and Computers
- Organizational Units
- User account management
- Security group management
- Domain joining

- Group Policy
- Network configuration
- Shared folder administration
- NTFS and share permissions
- Group-based access control
- Command-line troubleshooting
- Network connectivity testing
- DNS troubleshooting

The lab also provided practical experience diagnosing a DNS configuration issue and troubleshooting shared-folder permissions based on Active Directory security group membership.